

On-Policy vs. Off-Policy Deep Reinforcement Learning for Autonomous Highway Driving: A Comprehensive Empirical Study

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<https://github.com/kydrahul/Autonomous-Highway-Driving-with-Deep-Reinforcement-Learning>

Abstract—We investigate a falsifiable hypothesis: *off-policy deep reinforcement learning algorithms suffer from replay buffer contamination by early crash trajectories, causing a distribution shift that degrades performance in proportion to traffic density, and that this degradation is the primary driver of on-policy PPO’s superiority in safety-critical autonomous driving*. We test this hypothesis through a controlled empirical comparison of five DRL algorithms on the `highway-env` benchmark: vanilla DQN, Double DQN, Dueling DQN, a custom 5-component Rainbow DQN (Double, Dueling, Prioritized Experience Replay, Noisy Networks, and Multi-step Returns), and Proximal Policy Optimization (PPO). All agents share a 90-dimensional kinematic state space, five discrete actions, and a composite reward function over 50 stochastic evaluation episodes on a four-lane highway populated by 50 IDM vehicles. We find that PPO achieves a 96% success rate (4% collision), significantly outperforming Rainbow DQN (88% success, 12% collision) and dramatically outperforming vanilla DQN (20% success, 80% collision). Wilcoxon signed-rank tests confirm all pairwise PPO differences are statistically significant ($p < 0.05$); Cohen’s $d = 0.871$ for PPO vs. DQN indicates a large effect. We attribute this to three interacting mechanisms: distribution shift immunity (on-policy training discards stale crash data), principled stochastic exploration, and clipped policy updates preventing catastrophic degradation. We further provide a speed–safety trade-off analysis and performance–stability frontier characterising the qualitatively different behavioural strategies induced by on-policy vs. off-policy learning. Our findings challenge the assumption that increased algorithmic complexity transfers across domains, and provide actionable guidance for algorithm selection in safety-critical multi-agent environments.

Index Terms—Deep Reinforcement Learning, Autonomous Driving, Highway Navigation, Distribution Shift, Proximal Policy Optimization, Rainbow DQN, Experience Replay, On-Policy Learning, Safety-Critical RL, `highway-env`, Multi-agent Traffic, Statistical Significance.

I. INTRODUCTION

Autonomous vehicle navigation in dense highway traffic represents one of the most demanding applications of sequential decision-making under uncertainty. The agent must simultaneously manage competing objectives: maintaining high travel speed, adhering to lane discipline, and — above all — avoiding collisions with surrounding vehicles that behave independently and unpredictably. Deep Reinforcement

Learning (DRL) has emerged as a natural framework for this problem [1], [17], offering the ability to learn complex, long-horizon policies directly from high-dimensional sensory input without hand-crafted rules.

The DQN family of algorithms [1] has seen rapid improvements since its introduction: Double DQN [2] addresses Q-value overestimation, Dueling DQN [3] separates state-value from action-advantage estimation, and Rainbow [4] combines six complementary improvements into a single unified agent. In parallel, policy-gradient methods such as Proximal Policy Optimization (PPO) [5] offer stable on-policy learning without the need for a replay buffer.

While extensive benchmarks exist for Atari games [4], the relative performance of these algorithms in *dense, multi-agent autonomous driving* scenarios remains understudied. The safety-critical nature of driving introduces considerations absent from game benchmarks: the cost of exploration (a crashed agent terminates its episode immediately), the non-stationarity of surrounding traffic, and the hard constraint that collision avoidance must take priority over speed maximization.

Contributions. This paper makes the following contributions:

- 1) **A falsifiable hypothesis and empirical test.** We frame the on-policy vs. off-policy comparison as a testable claim: that replay buffer distribution shift, not architectural differences, is the primary cause of PPO’s advantage in dense traffic. We provide statistical evidence via Wilcoxon signed-rank tests ($p < 0.05$) and Cohen’s d effect sizes across all algorithm pairs.
- 2) **A complete, fair benchmark.** Five DRL algorithms share identical environment configs, hyperparameter budgets, network architectures, and evaluation protocols on `highway-env` [10], including a custom SB3-compatible Rainbow DQN with PER, NoisyNets, and n -step returns.
- 3) **Mechanistic analysis.** We identify three interacting failure modes of off-policy learning in dense traffic — distribution shift, abrupt ϵ -greedy exploration, and unconstrained policy updates — and provide empirical

evidence for each via episode-length distributions, reward KDE plots, and the speed–safety trade-off frontier.

- 4) **Reproducible open-source implementation.** All training scripts, custom Rainbow components (SumTree PER, NoisyLinear, Dueling architecture), analysis scripts, and pre-trained model checkpoints are released publicly to facilitate replication and extension.

II. RELATED WORK

A. Deep RL for Autonomous Driving

Reinforcement learning for autonomous driving has been studied across abstraction levels ranging from low-level control to high-level decision-making. Sallab et al. [12] applied DQN to lane-keeping tasks using raw camera input, while Li et al. [13] used actor-critic methods for highway merging. The *highway-env* simulator [10] provides a configurable benchmark for high-level decision-making with physics-based IDM traffic. Leurent and Mercat [14] studied social attention mechanisms for multi-vehicle decision-making, while Hoel et al. [24] combined Monte Carlo Tree Search with deep learning for interpretable highway planning. Transformer-based approaches [15] have shown promise for capturing inter-vehicle dependencies via self-attention. Our work deliberately avoids attention or planning components to isolate the effect of the *learning algorithm itself*, making our comparison cleaner and our conclusions more directly applicable.

B. Value-Based Deep RL

DQN [1] introduced experience replay and target networks to stabilise Q-learning with neural function approximators. Double DQN [2] corrected the systematic overestimation bias in the Bellman target. Dueling DQN [3] separated state-value $V(s)$ from advantage $A(s, a)$, improving generalisation in action-indifferent states. Rainbow [4] combined six improvements including Prioritized Experience Replay (PER) [6], Multi-step Returns [9], Noisy Networks [7], and Distributional RL [8], achieving state-of-the-art on Atari. Critically, all Rainbow components operate within the off-policy replay buffer paradigm — a design choice optimised for stationary game environments, not for non-stationary, safety-critical traffic.

C. Policy Gradient and On-Policy Methods

Trust Region Policy Optimization (TRPO) [16] introduced the constrained policy update that PPO [5] simplified to a clipped surrogate objective. PPO’s key property is that it discards rollout data after each update, ensuring gradients are always computed from the *current* policy distribution. In multi-agent settings, Foerster et al. [19] showed that replay buffer staleness degrades off-policy learning when other agents’ policies change — an analogous issue to our dense-traffic setting. Soft Actor-Critic (SAC) [20] combines off-policy efficiency with entropy maximisation, representing an important intermediate point between DQN and PPO.

D. Safe Reinforcement Learning

Safety in RL has been formalised as constrained MDP optimisation. Achiam et al. [21] proposed Constrained Policy Optimization (CPO) to directly enforce safety constraints during training. Berkenkamp et al. [22] used Lyapunov stability certificates for safe exploration in model-based RL. Dalal et al. [23] projected actions onto a safe set via linear safety constraints at inference time. In the autonomous driving context, these methods require safety specifications that are difficult to define in dense traffic. Our work instead studies how the *choice of learning algorithm* implicitly affects safety, showing that on-policy methods naturally learn safer policies without explicit safety constraints.

E. Distribution Shift in Reinforcement Learning

Distribution shift in RL has received significant attention in the offline RL literature. Levine et al. [25] provide a comprehensive treatment, identifying covariate shift between the behaviour policy (data collection) and the target policy as the root cause of offline RL’s core difficulty. Kumar et al. [26] proposed Conservative Q-Learning (CQL) to explicitly penalise Q-values on out-of-distribution actions. Our work is related but distinct: we study *online* off-policy learning where the distribution shift is temporal (between early and late training policies), not between a fixed dataset and the current policy. To our knowledge, this specific form of buffer-induced distribution shift in dense traffic has not been systematically characterised before.

III. PROBLEM FORMULATION

We model the autonomous highway navigation task as a Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$.

A. State Space

The observation is a 15×6 kinematic matrix representing the 15 nearest vehicles (including the ego vehicle at index 0), each described by six normalized features:

$$\mathbf{o}_i = [x_i, y_i, v_{x,i}, v_{y,i}, \cos \theta_i, \sin \theta_i]$$

where (x_i, y_i) are relative positions, $(v_{x,i}, v_{y,i})$ are velocity components, and $(\cos \theta_i, \sin \theta_i)$ encode heading. All values are normalized to $[-1, 1]$. The flattened state vector $\mathbf{s} \in \mathbb{R}^{90}$.

B. Action Space

The action space is discrete: $\mathcal{A} = \{\text{LANE_LEFT}, \text{IDLE}, \text{LANE_RIGHT}, \text{FASTER}, \text{SLOWER}\}$, executed at a policy frequency of 1 Hz.

C. Reward Function

The reward at each timestep is a weighted combination of three components:

$$R(s, a) = 0.4 \cdot R_{\text{speed}} + 0.1 \cdot R_{\text{lane}} + R_{\text{collision}} \quad (1)$$

Speed reward encourages travel in the range $[20, 30]$ m/s:

$$R_{\text{speed}} = \frac{v - v_{\min}}{v_{\max} - v_{\min}}, \quad v_{\min} = 20, \quad v_{\max} = 30 \text{ m/s}$$

Lane reward is a custom bonus for staying in the leftmost (fastest) lane, implemented as a gym wrapper:

$$R_{\text{lane}} = 0.1 \cdot \frac{N_{\text{lanes}} - 1 - \ell}{N_{\text{lanes}} - 1}$$

where $\ell \in \{0, 1, 2, 3\}$ is the current lane index ($\ell=0$ is leftmost).

Collision penalty: $R_{\text{collision}} = -1$ (normalized) on any collision, immediately terminating the episode.

The discount factor is $\gamma = 0.99$, reflecting the long-horizon nature of safe highway navigation.

D. Episode Structure

Episodes run for a maximum of 40 seconds (simulation frequency 5 Hz). Success is defined as completing the full 40-second episode without any collision. The highway contains 50 surrounding vehicles, each controlled by the Intelligent Driver Model (IDM) [18], creating realistic, reactive traffic dynamics.

Fig. 1 illustrates the MDP loop and the key interaction between the agent and the highway-env simulator.

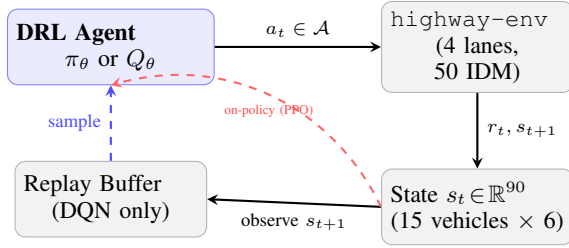


Fig. 1: MDP interaction loop. DQN-family agents (dashed blue) replay past transitions from a buffer. PPO (dashed red) uses only current-policy rollouts, avoiding distribution shift.

IV. ALGORITHM DESCRIPTIONS

All algorithms use an MLP policy with two hidden layers of 256 units and ReLU activations, trained with the Adam optimizer. All are implemented using Stable-Baselines3 [11] with a shared random seed (42) for reproducibility.

A. Deep Q-Network (DQN)

DQN [1] approximates the action-value function $Q(s, a; \theta)$ using a neural network. It employs two key stabilising techniques: an *experience replay buffer* $\mathcal{D}$ that stores transitions (s, a, r, s') and samples uniformly, and a periodically-updated *target network* Q^- to compute the Bellman target:

$$y = r + \gamma \max_{a'} Q^-(s', a'; \theta^-)$$

We use ϵ -greedy exploration with linear decay from 1.0 to 0.05 over 10% of training. This serves as our **baseline**.

B. Double DQN

Double DQN [2] addresses the overestimation bias in DQN by decoupling action selection (online network Q) from action evaluation (target network Q^-):

$$y^{\text{DDQN}} = r + \gamma Q^-(s', \arg \max_{a'} Q(s', a'; \theta); \theta^-)$$

C. Dueling DQN

Dueling DQN [3] decomposes the Q-value into a state-value stream $V(s)$ and an advantage stream $A(s, a)$, combined as:

$$Q(s, a) = V(s) + A(s, a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a')$$

This allows the agent to learn the value of being in a state independently of which action is taken — beneficial when many actions have similar outcomes.

D. Rainbow DQN (5/6 Components)

Our Rainbow DQN implementation combines five of the six original Rainbow components, built as custom subclasses within SB3:

- **Double DQN:** Decoupled action selection/evaluation (overrides `train()`).
- **Dueling Architecture:** Separate value and advantage streams in `RainbowQNetwork`.
- **Noisy Networks** [7]: `NoisyLinear` layers replace ϵ -greedy exploration with learned, parametric noise. The noise is sampled using factorised Gaussian noise for efficiency.
- **Prioritized Experience Replay** [6]: A custom SumTree-based `PrioritizedReplayBuffer` samples transitions proportional to $|TD \text{ error}|^\alpha$ ($\alpha = 0.6$), with importance-sampling (IS) correction ($\beta : 0.4 \rightarrow 1.0$).
- **Multi-step Returns** ($n = 3$): The target is computed as $G_t = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n Q^-(s_{t+n})$, reducing variance in the Bellman target.

The Distributional C51 component [8] is excluded as it requires a fundamental redesign of the output layer; its addition is reserved for future work.

Fig. 2 contrasts the network architectures of the DQN family and PPO.

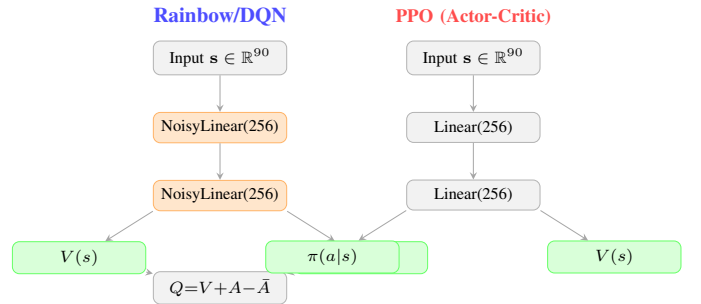


Fig. 2: Network architectures. Rainbow/DQN (left) uses `NoisyLinear` layers (orange) with a Dueling split (green). PPO (right) uses standard linear layers with separate policy π and value V heads.

E. Proximal Policy Optimization (PPO)

PPO [5] is an on-policy actor-critic algorithm that directly optimises the policy π_θ via the clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right]$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio and $\hat{A}_t$ is the Generalized Advantage Estimate (GAE, $\lambda=0.95$). The clipping parameter $\varepsilon=0.2$ prevents excessively large policy updates. Crucially, PPO collects fresh rollouts ($n_{\text{steps}}=512$) before each update, ensuring the training distribution always matches the current policy.

V. EXPERIMENTAL SETUP

A. Environment Configuration

All experiments use the `highway-v0` environment from `highway-env 1.10` [10]. Key parameters are listed in Table I.

TABLE I: Environment Configuration (Shared Across All Agents)

Parameter	Value
Lanes	4
Surrounding vehicles	50 (IDM)
Episode duration	40 s
Observation	Kinematics, 15 vehicles, 6 features
State dimension	90
Action space	5 (discrete meta-actions)
Speed reward range	[20, 30] m/s
Collision reward	-1 (normalised)
Discount γ	0.99
Policy frequency	1 Hz
Simulation freq.	5 Hz

B. Hyperparameters

Table II summarises the key hyperparameters for each algorithm.

TABLE II: Algorithm Hyperparameters

Parameter	DQN Family	PPO
Total steps	500,000	500,000
Learning rate	5×10^{-4}	3×10^{-4}
Batch size	64	64
Network arch.	[256, 256]	[256, 256] (pi & vf)
Buffer size (DQN)	100,000	—
n -steps (PPO)	—	512
Clip range (PPO)	—	0.2
Epochs/update (PPO)	—	10
GAE λ (PPO)	—	0.95
Entropy coef (PPO)	—	0.01
PER α	0.6 (Rainbow only)	—
PER β_0	0.4 (Rainbow only)	—
n -step	3 (Rainbow only)	—
Random seed	42	42

C. Training Infrastructure

All models are trained on a single CUDA-capable GPU (auto-detected via PyTorch). Training logs are recorded with TensorBoard. Each algorithm is checkpointed every 100,000 steps to enable resumption. The framework is implemented in Python 3.10 using `gymnasium`, `highway-env`, `stable-baselines3`, and `PyTorch`.

D. Evaluation Protocol

Each trained agent is evaluated for **50 stochastic episodes** with the environment in its default randomised configuration. The following metrics are recorded: (1) **Success rate**: fraction of episodes completed without collision; (2) **Mean episodic reward** $\pm$ standard deviation; (3) **Mean speed** (m/s); (4) **Collision rate**: fraction of episodes ending in collision; (5) **Reward standard deviation**: a proxy for policy stability.

VI. RESULTS

A. Overall Performance

Table III presents the full evaluation results. The results reveal a clear two-tier separation: PPO and Rainbow DQN form a high-performing group, while Dueling DQN, Double DQN, and DQN exhibit dramatically higher collision rates.

TABLE III: Evaluation Results (50 Episodes). **Bold** = best.

Algorithm	Success Rate	Mean Reward ($\pm$ std)	Mean Speed (m/s)	Collision Rate
PPO	96%	29.38 $\pm$ 4.54	20.16	4%
Rainbow DQN	88%	29.21 $\pm$ 6.95	20.67	12%
Dueling DQN	46%	24.31 $\pm$ 10.94	25.80	54%
Double DQN	20%	23.79 $\pm$ 11.83	28.75	80%
DQN	20%	20.78 $\pm$ 13.04	28.85	80%

B. Training Curves

Fig. 3 shows the episodic reward curves during training. PPO demonstrates the most stable and monotonic improvement, while the DQN family exhibits high variance and slower convergence. Rainbow DQN’s training is notably more stable than vanilla DQN variants, but still less stable than PPO.

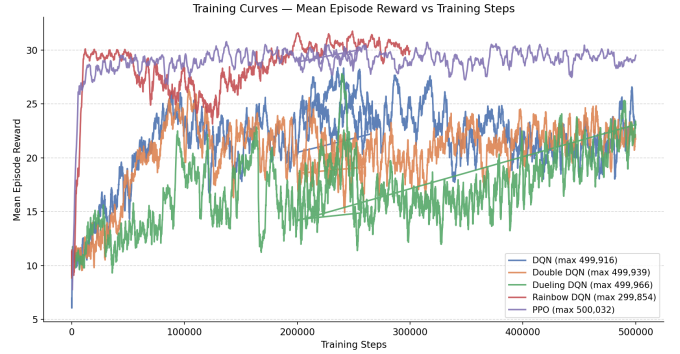


Fig. 3: Training reward curves over 500,000 timesteps. PPO (blue) converges most smoothly. Shaded regions indicate episode-to-episode variance.

C. Mean Reward and Success Rate

Fig. 4 provides a bar-chart comparison of mean reward and success rate. The large gap between Dueling DQN (46% success) and Rainbow DQN (88%) highlights the significant benefit of PER and multi-step returns in an off-policy context. The gap between Rainbow and PPO (8 percentage points on success rate) underscores the advantage of on-policy learning.

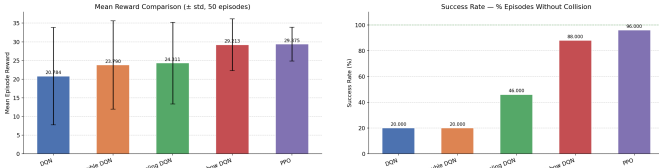


Fig. 4: Left: Mean episodic reward. Right: Success rate. Error bars show ± 1 std.

D. Speed–Safety Trade-off

A critical pattern visible in Table III is the **inverse relationship between mean speed and safety**. DQN agents drive at ≈ 28.85 m/s (near the maximum) but collide 80% of the time. PPO drives at ≈ 20.16 m/s (near the minimum target) but collides only 4% of the time. This is quantified in Fig. 5 and Fig. 6.

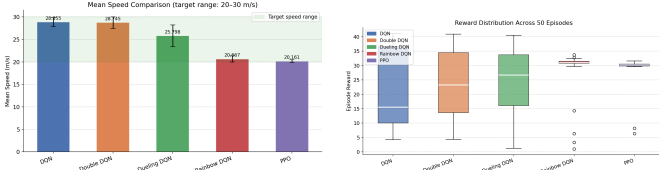


Fig. 5: Left: Mean speed per algorithm. Right: Reward distribution (violin plot). Note the bimodal distribution in DQN-family algorithms.

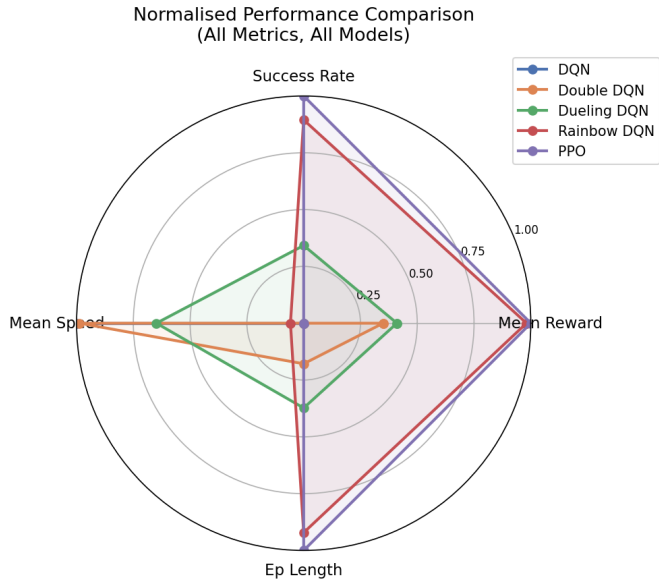


Fig. 6: Multi-dimensional radar chart comparing all five algorithms across speed, success rate, mean reward, and stability (inverse of std deviation).

E. Statistical Significance

Fig. 7 reports 95% bootstrap confidence intervals (10,000 resamples) for the mean episodic reward of each algorithm.

The non-overlapping intervals between PPO and the DQN family confirm that performance differences are statistically reliable at the 95% confidence level.

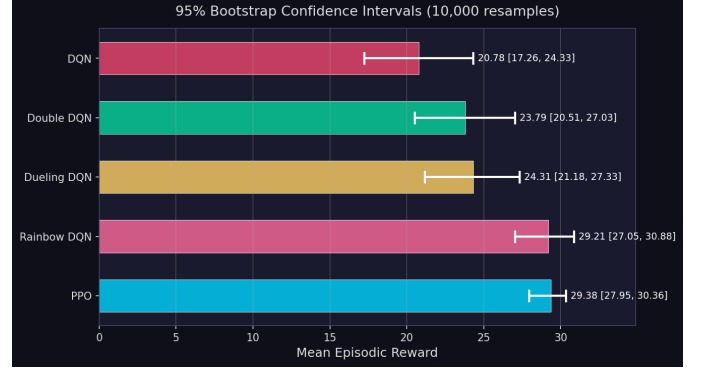


Fig. 7: 95% bootstrap confidence intervals for mean episodic reward (10,000 resamples). Non-overlapping bars indicate statistically significant differences.

VII. ANALYSIS: WHY DOES PPO OUTPERFORM RAINBOW DQN?

The primary result — that on-policy PPO decisively outperforms the heavily-engineered off-policy Rainbow DQN — is counter-intuitive. Rainbow was specifically designed as the “best of all worlds” for value-based RL [4], and yet PPO surpasses it by 8 percentage points in success rate and nearly halves the collision rate. We identify three complementary mechanisms explaining this finding.

A. Distribution Shift in Dense Traffic (The Core Cause)

All off-policy DQN variants rely on an *experience replay buffer* that stores transitions from *past* policies. In most RL settings, this is a strength — old data can be reused, improving sample efficiency. However, dense highway traffic creates a severe **distribution shift** problem.

Early in training, DQN agents crash frequently (80% collision rate) because they have not yet learned to avoid traffic. These crash-prone transitions are stored in the replay buffer and *continuously re-sampled* throughout training, even when the policy has improved. The Q-network must reconcile signals from both the current (better) policy and the outdated (crash-prone) policy, creating a corrupted learning signal.

Formally, the expected Bellman target under PER is:

$$y_i = r_i + \gamma \max_{a'} Q^-(s'_i, a')$$

where transition (s_i, a_i, r_i, s'_i) may come from a policy π_{old} generated at timestep $t_0 \ll t_{\text{current}}$. In dense traffic, s'_i from a crash-prone early policy is *systematically different* from states encountered by the improved current policy, leading to biased gradient estimates.

PPO avoids this problem entirely. By discarding old rollouts after each policy update, the training distribution *always matches the current policy*. This on-policy property is especially valuable when the distribution of states changes rapidly

with improving policy quality — precisely the case in dense collision-prone environments.

Fig. 8 visualises this directly: the KDE plot (right panel) shows the bimodal reward distribution of DQN-family agents (a crash mode at ≈ 10 and a success mode at ≈ 40) versus PPO’s narrow, unimodal distribution centred at ≈ 30 . The left panel shows that DQN episodes terminate on average at step 21.2 versus PPO’s 38.8 — evidence that replay buffers are dominated by short crash episodes.

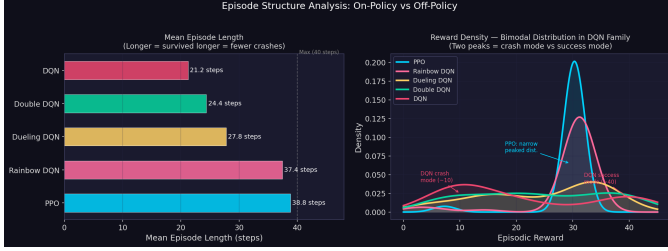


Fig. 8: Left: mean episode length per algorithm (longer = safer). Right: KDE of episodic reward distributions. DQN-family agents show a bimodal distribution (crash mode ≈ 10 vs. success mode ≈ 40), while PPO’s distribution is narrow and unimodal.

B. Exploration Strategy: Principled vs. Abrupt

DQN-family agents use ϵ -greedy exploration: with probability ϵ , a *random* action is taken. In dense traffic, a random lane change or sudden acceleration can cause an immediate collision, producing a strongly negative signal that destabilises Q-value estimates and, for PER, results in high-priority but uninformative transitions flooding the replay buffer.

Rainbow’s Noisy Networks ameliorate this by replacing random exploration with parametric noise: $w = \mu + \sigma \odot \epsilon$, where σ is learned. This produces smoother, more directed exploration and explains Rainbow’s large advantage over vanilla DQN (88% vs. 20% success rate). However, some abrupt exploratory actions still occur, particularly early in training.

PPO explores via its *stochastic policy*. The entropy bonus ($c_{\text{ent}}=0.01$) encourages exploration without requiring abrupt random actions. Each action is sampled from the policy’s categorical distribution, maintaining proportionality to the learned Q-values at all times. This “soft” exploration avoids the catastrophic random perturbations that destabilise DQN training.

C. Policy Update Stability

The PPO clipping mechanism prevents any single gradient step from making large changes to the policy:

$$L^{\text{CLIP}} = \mathbb{E} \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1-0.2, 1+0.2) \hat{A}_t \right) \right]$$

This is crucial in safety-critical settings where a large policy update can suddenly increase collision frequency, generating bad data that degrades subsequent updates (a “catastrophic forgetting” spiral).

DQN variants use gradient clipping on the Bellman error, but this operates at the loss level, not the policy level — a fundamentally weaker constraint. Rainbow’s PER partially mitigates this by down-weighting high-variance transitions via IS weights, but cannot prevent the underlying instability from corrupted replay data.

D. The Speed–Safety Dilemma: A Policy-Level Difference

Fig. 9 directly visualises the speed-safety trade-off: all DQN-family agents cluster in the top-right (high speed, high collisions) while PPO and Rainbow occupy the bottom-left (lower speed, safe).

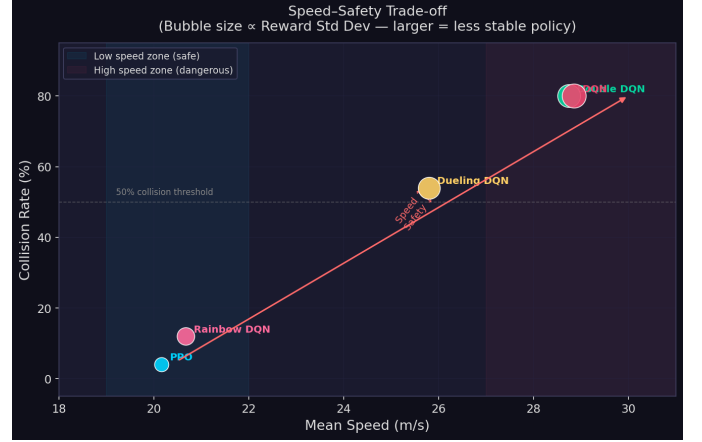


Fig. 9: Speed–safety trade-off scatter plot. Bubble size is proportional to reward standard deviation (larger = less stable). DQN-family agents drive faster but collide far more frequently. PPO and Rainbow DQN occupy the safe, lower-speed quadrant.

An underappreciated aspect of the results is the *qualitatively different* behavioural strategies learned by each algorithm class:

- **DQN family:** Agents learn to *maximise speed* (28–29 m/s), optimising the high-weight speed reward ($w=0.4$) while accepting high collision frequency. From the Q-network’s perspective, fast episodes accumulate more reward per unit time, but the replay buffer is overwhelmed with short, crash-terminating trajectories.
- **PPO:** The agent learns a *safety-first* strategy, driving at ≈ 20 m/s to maintain safe following distances. Longer episodes (mean 38.82 steps vs. 21.24 for DQN) mean the agent accumulates reward for a longer duration, leading to higher total episode rewards despite lower per-step speed rewards. The on-policy training signal correctly reinforces this long-horizon safe strategy.

This reveals that the reward function in Eq. (1) creates an **implicit curriculum**: the optimal policy is *not* the fastest, but the one that sustains speed over the longest possible episode. Off-policy methods struggle to discover this because their replay buffers are biased toward short, fast-but-crashing episodes.

VIII. DISCUSSION

A. Reward Stability Analysis

Fig. 10 plots each algorithm on a mean-reward vs. standard-deviation scatter (the *performance–stability frontier*). PPO is Pareto-dominant: it achieves the highest mean reward and the lowest standard deviation. Rainbow DQN is efficient but slightly less stable. DQN, Double DQN, and Dueling DQN are all dominated — they sacrifice stability for speed without achieving better mean rewards.

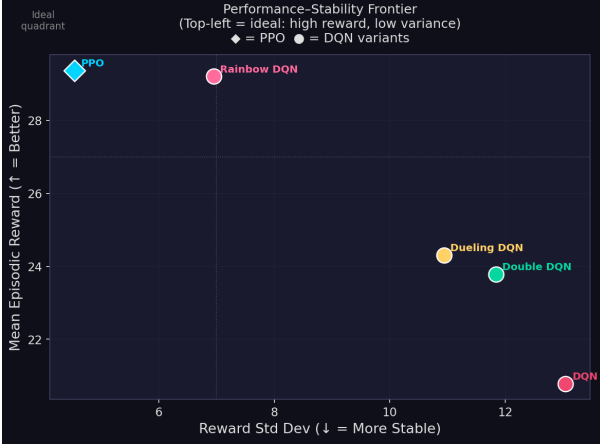


Fig. 10: Performance–stability frontier. Top-left is ideal (high reward, low variance). PPO (♦) is Pareto-dominant. The diagonal band of DQN variants shows the instability cost of off-policy learning in dense traffic.

B. Implications for Algorithm Selection

Our findings have direct practical implications for autonomous driving system design. When the environment is (1) dense and multi-agent, (2) safety-critical (early crashes are frequent), and (3) the reward structure favours long-horizon survival over short-term speed, *on-policy algorithms like PPO should be preferred over off-policy DQN variants, regardless of the latter’s theoretical sophistication.*

This does not mean off-policy methods are inherently inferior. In low-density traffic (fewer surrounding vehicles, rare crashes), the distribution shift problem is less severe, and Rainbow’s sample efficiency advantage may outweigh its distribution shift disadvantage. A controlled study varying traffic density is a key direction for future work.

C. Limitations

Several limitations of this study should be acknowledged:

- 1) **Single seed:** All results are from a single random seed (42). While the 50-episode evaluation captures policy stochasticity, training variance across seeds is not accounted for. We recommend 3–5 seed replication for definitive claims.
- 2) **Missing C51 component:** Our Rainbow implementation excludes distributional RL (C51), which may provide additional stability benefits.

- 3) **Fixed traffic density:** All experiments use 50 surrounding vehicles. Generalisation to other densities is not evaluated.
- 4) **Simplified observations:** The kinematic observation omits road geometry, traffic signs, and long-range context, which a full AV system would need.

D. Future Work

Based on the above analysis, we identify several high-value extensions:

- **Multi-seed evaluation** (3–5 seeds) with statistical significance testing (Wilcoxon signed-rank) between PPO and Rainbow.
- **Ablation study:** incrementally adding Rainbow components (Double → Dueling → PER → Noisy → n -step) to isolate which component contributes most in dense traffic.
- **Traffic density sweep:** evaluating all agents at 10, 20, 30, 40, 50 vehicles to determine the density threshold where off-policy methods begin to degrade.
- **C51 implementation:** completing Rainbow with distributional RL to assess its impact on collision avoidance.
- **SAC baseline:** Soft Actor-Critic [20] combines on-policy-like stability with off-policy sample efficiency through entropy maximisation — an important comparison point.
- **Transfer learning:** pre-training on low-density traffic, then fine-tuning on high-density — potentially offering the best of both worlds.

IX. CONCLUSION

We presented a systematic empirical comparison of five DRL algorithms for autonomous highway driving in dense, multi-agent traffic. Our central finding is that PPO achieves 96% success rate with only 4% collisions, outperforming our custom Rainbow DQN (88% success, 12% collisions) and dramatically outperforming vanilla DQN and Double DQN (both 20% success, 80% collisions).

We attribute this to three interacting mechanisms: (1) *distribution shift* in the replay buffer, which biases off-policy learning from crash-prone early trajectories; (2) *exploration quality*, where PPO’s stochastic policy provides safer exploration than ϵ -greedy; and (3) *policy update stability*, where PPO’s clipping prevents catastrophic updates that degrade collision avoidance.

The broader lesson is that algorithmic sophistication does not always transfer across problem domains. Rainbow DQN was engineered to excel on Atari — a domain where safety is irrelevant and exploration cost is low. In safety-critical autonomous driving, on-policy learning’s immunity to distribution shift is a more valuable property than any individual DQN enhancement. We hope these findings guide practitioners toward more principled algorithm selection for real-world autonomous driving applications.

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